

# Yige Yan

Ph.D. Candidate, College of Computing and Data Science, Nanyang Technological University

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## Research Profile

My research focuses on **pathology–age decoupling** for neurodegenerative diseases in brain functional connectivity and **contrast-free microvascular obstruction identification** from cine cardiac MRI. Seeking research internships (medical AI, multimodal learning, LLMs, AI agents, computer vision) before thesis submission in the first half of 2027, and postdoctoral positions from 2027.

## Education

**Ph.D.**, Computer Science and Engineering, Nanyang Technological University *Aug 2023 – Present*  
Supervisors: Prof. Jagath C. Rajapakse (IEEE Fellow) and Assoc. Prof. Nagaendran Kandiah.

Expected thesis submission: first half of 2027.

**M.Sc.**, Computer Control and Automation, Nanyang Technological University *Aug 2022 – Jun 2023*

**B.Eng.**, Computer Science, Hohai University, China *Sep 2017 – Jun 2021*

## Publications & Manuscripts

- Yige Yan**, Yi Hao Chan, Jun Cheng, Qian Hui Chew, Kang Sim, and Jagath C. Rajapakse. “Latent Pathology–Age Decoupling in Functional Connectivity Graphs for Multi-task Schizophrenia Assessment.” *MICCAI*, 2026. **Early Accept.**
- Yige Yan**, Jun Cheng, Xulei Yang, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Dual Correlation-Aware Mamba for Microvascular Obstruction Identification in Non-contrast Cine Cardiac Magnetic Resonance.” *MICCAI*, 2025. **Early Accept.**
- Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cardiac Magnetic Resonance.” *MICCAI*, 2024. **Early Accept.**
- Yige Yan**, Jing Xia, Yi Hao Chan, Thomas Welton, and Jagath C. Rajapakse. “Decoupling Age Effects from Parkinson’s Disease Functional Connectivity.” *Medical Image Analysis*. **Under review.**
- Yige Yan**, Jun Cheng, Xulei Yang, Zaiwang Gu, Shuang Leng, Ru San Tan, Liang Zhong, and Jagath C. Rajapakse. “Spatiotemporal Adaptive Mamba with Coarse-Grained Mask Regularization for Microvascular Obstruction Identification from Non-contrast Cine Cardiac MRI.” *Neurocomputing*. **Under review.**
- Yige Yan**, Jing Xia, Yi Hao Chan, Nagaendran Kandiah, and Jagath C. Rajapakse, for the Alzheimer’s Disease Neuroimaging Initiative. “Clinical Evidence Reconfigures Age Reliance in Language Models Assessing Alzheimer’s Disease.” *Target: npj Digital Medicine*.
- Yige Yan**, Pei Yang, Jun Cheng, Lianggeng Gong, and Jagath C. Rajapakse. “Temporal Variation Proxy and Staged Proposal-Support Integration for Contrast-Free MVO Segmentation in Cine CMR.” *Target: IEEE Transactions on Medical Imaging*.
- Yige Yan**, Hang Wang, Yi Hao Chan, and Jagath C. Rajapakse. “Auditing Named Knowledge Interfaces: Regularization, Responsiveness, and Predictive Utility.” *Target: Knowledge-Based Systems*.

## Research Experience

**Nanyang Technological University** Singapore  
*Ph.D. Researcher, College of Computing and Data Science* *Aug 2023 – Present*

- Developed **contrast-free microvascular obstruction identification** from cine cardiac MRI.
- Developed **pathology–age decoupling** in brain functional connectivity for schizophrenia, Parkinson’s and Alzheimer’s disease.

**Institute for Infocomm Research (I<sup>2</sup>R), A\*STAR** Singapore  
*Research Student Attachment, Medical Imaging* *Aug 2023 – Present*  
Mentor: Dr. Jun Cheng.

- Microvascular obstruction identification from non-contrast cine cardiac MRI, with collaborators across A\*STAR, NTU, Duke-NUS Medical School and the National Heart Centre Singapore.

Full CV: [code-koukai.github.io/Yige\\_Yan\\_CV\\_Full.pdf](https://code-koukai.github.io/Yige_Yan_CV_Full.pdf)